Basic microcontroller (msp430) programming using C language

LAB # 02

Spring 2026

CSE-307L Microprocessor-Based System Design Lab

Submitted by: **AKMAL SULTAN**

Registration No: **23PWCSE2252**

Class Section: **B**

Submitted to:

Engr: Shahzada Fahim Jan

(February 16, 2026)

Department of Computer Systems Engineering

University of Engineering and Technology, Peshawar

TASKS

TASK 1

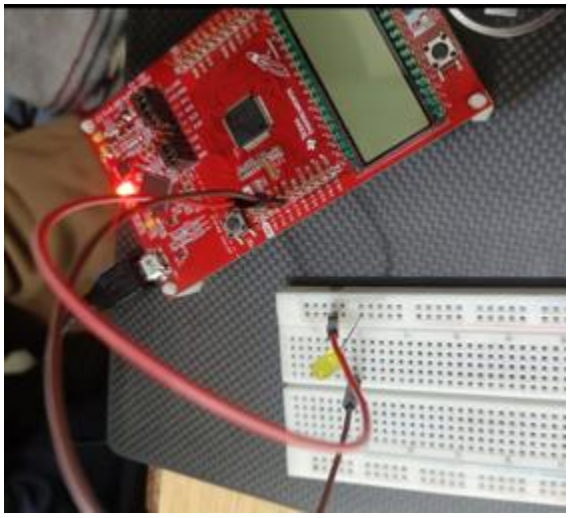
Write C program for Msp430 which toggle P1.0 or any other Pin of Msp430 MCU.

Code:

emptyProject > main.c > ...

```c
#include <msp430.h>

void delay(void);

int main(void)
{
    WDTCTL = WDTPW | WDTHOLD;   // Stop watchdog timer

    P1DIR |= BIT0;              // Set P1.0 as output

    while(1)
    {
        P1OUT ^= BIT0;          // Toggle P1.0
        delay();
    }
}

void delay(void)
{
    unsigned int i;
    for(i = 0; i < 50000; i++);
}
```

Output:



TASK2:

Write C program which toggle the LEDS attached with P1.0 and P1.7 at the same time with different delays.

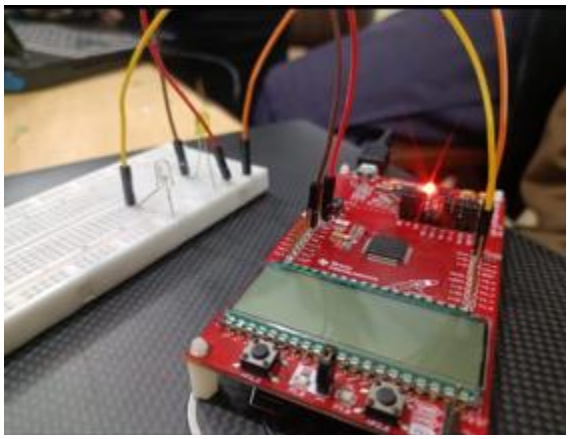
Code

Get Started | Project Wizard | main.c

emptyProject > main.c > ...

```c
#include <msp430.h>

void delay1(void);
void delay2(void);

int main(void)
{
    WDTCTL = WDTPW | WDTHOLD;
    P1DIR |= BIT0 + BIT7;   // Set P1.0 and P1.7 as output
    while(1)
    {
        P1OUT ^= BIT0;      // Toggle LED at P1.0
        delay1();
        P1OUT ^= BIT7;      // Toggle LED at P1.7
        delay2();
    }
}
void delay1(void)
{
    unsigned int i;
    for(i = 0; i < 30000; i++);
}
void delay2(void)
{
    unsigned int i;
    for(i = 0; i < 70000; i++);
}
```

Output:



TASK3:

Write C program which toggle all the LEDs attached with P1 or any other PORT.

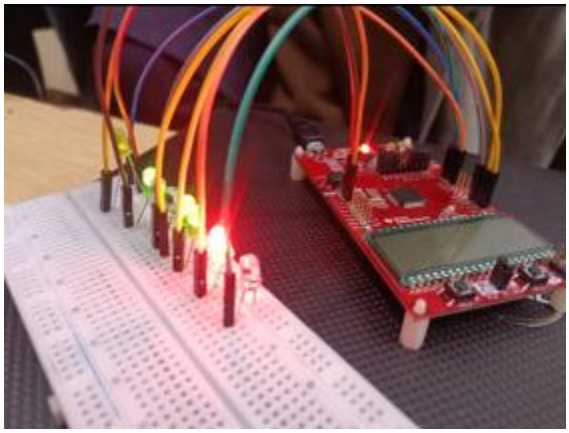
Code:

Get Started | Project Wizard | main.c

emptyProject > main.c > ...

```c
#include <msp430.h>

void delay(void);

int main(void)
{
    WDTCTL = WDTPW | WDTHOLD;

    P1DIR = 0xFF;    // Set all P1 pins as output

    while(1)
    {
        P1OUT ^= 0xFF;   // Toggle all bits of Port 1
        delay();
    }
}

void delay(void)
{
    unsigned int i;
    for(i = 0; i < 50000; i++);
}
```

Output:



TASK4:

Display the pattern using C language

00000001

00000010

00000100

....

10000000

00000001

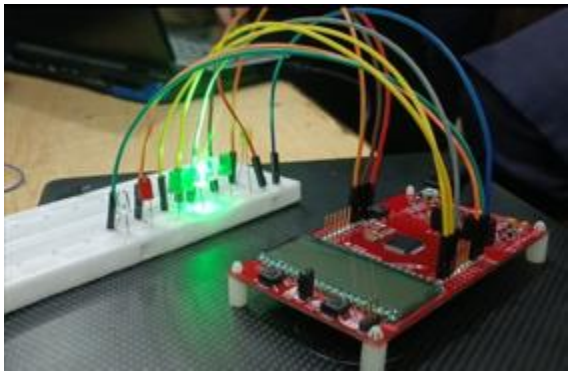
00000010

Continuously

Code:

Get Started | Project Wizard | main.c

emptyProject > main.c > ...

```c
#include <msp430.h>

void delay(void);

int main(void)
{
    WDTCTL = WDTPW | WDTHOLD;

    P1DIR = 0xFF;      // Port 1 as output
    unsigned char pattern = 0x01;

    while(1)
    {
        P1OUT = pattern;   // Output pattern to LEDs
        delay();

        pattern = pattern << 1;   // Shift left

        if(pattern == 0x00)
            pattern = 0x01;       // Restart from 00000001
    }
}

void delay(void)
{
    unsigned int i;
    for(i = 0; i < 50000; i++);
}
```

Output: